Bryson Wingate

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https://github.com/SuperiorPython | https://bryson-win-portfolio.netlify.app

Education

North Carolina A&T State University — Greensboro, NC
Bachelor of Computer Science — August 2023 - May 2027
- GPA: 3.85

Google Tech Exchange — Greensboro, NC
Technical Program — Jan 2025 - May 2025

Experience

Expedia Group — May 2026 – July 2026
Cloud Engineering Intern — San Jose, CA
- Built and deployed a production MCP server on Kubernetes exposing 5 custom observability tools over a 23.7 billion-row OTel ClickHouse dataset, enabling natural language querying of telemetry data.
- Designed query guardrails blocking 12+ dangerous SQL operations (DDL/DML) with automatic LIMIT injection and time-range enforcement, preventing unbounded queries against production data.
- Built a K6 performance test framework across 6 load scenarios (baseline, spike, soak, multi-tool) with JWT bearer auth, sustaining 133 requests/sec at 230ms p95 latency with zero errors across all scenarios.

CodePath / North Carolina A&T State University — Aug 2025 – Dec 2025 — Aug 2026 – Present
Tech Fellow — Greensboro, NC
- Mentored 150 students in Python and core CS concepts, increasing lab engagement and assignment completion rates in 42 lectures and led 23 labs
- Improved debugging proficiency and assignment completion through live code walkthroughs and structured feedback.
- Established responsible AI-use guidelines in coursework, reinforcing academic integrity and critical evaluation of generative outputs

Cisco — May 2025 – August 2025
Software Engineering Intern — Durham, NC
- Built and delivered an AI-powered chatbot prototype integrating Jira, GitHub, and Confluence to streamline ticket and documentation workflows
- Designed and deployed scalable AWS infrastructure (EC2 Auto Scaling, Load Balancers, VPC networking) for production-ready performance
- Presented architecture and projected impact to senior leadership, demonstrating a 15% reduction in documentation search time.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, HCL, C#, C++, SQL
Frameworks & Tools: React, Express, Docker, Kubernetes, AWS, Google Cloud, GitHub Actions, Unity, Streamlit, Git/GitHub

Projects

Astrolabe – NASA Space Dashboard | React, TypeScript, Express, NASA APIs — October 2026
- Built an interactive space dashboard unifying five NASA data feeds (APOD, ISS tracking, near-Earth objects, Exoplanet Archive, DONKI space weather), navigated from a draggable 3D globe.
- Developed an Express API layer that proxies and caches NASA endpoints to stay within rate limits, with a fallback ISS data source and custom Recharts and Leaflet visualizations.

Playlist DNA – Spotify Visualization App | Python, Streamlit, Spotipy, OpenAI API — October 2025
- Designed a data analysis pipeline using Spotipy to process 100+ playlists, performing genre frequency analysis and popularity trend modeling with interactive Altair visualizations.
- Integrated GPT-4o-mini for automated playlist summaries, implementing caching and rate-limit handling to improve load performance by ∼40% and ensure stable cloud deployment.